

Project Overview

Purpose:

The Redesign Project has two main objectives:

1. Apply the class design guidelines using software that is taught in class.
2. Present the results of the redesigns.
3. Additional: Build up your portfolio.

Submission Details:

Before **3:30 PM on March 16th, 2026**, you need to submit following:

1. URL of your Website with the Redesign Project.
2. URL of your GitHub Repository, that contains your website.
3. 10 – Min Video Presentation must be embedded to your website.
4. Codes of your Project must be included to your website as a **separate Webpage**.

No changes should be made to your GitHub repository after the mid project deadline and until your projects are graded.

Instructions:

You are required to identify an **already established visualization/s** from the web or another published source and create several redesigned versions of that same visualization. The goal is to critique and improve an existing visual—not to start with a dataset and create an intentionally poor graph first.

This process should help you uncover patterns or insights that were not immediately evident in the original design.

You are also expected to compare and critically evaluate the strengths and weaknesses of both the original visualization and your redesigns, taking the context into account.

The redesign project includes updating your website with the redesigned work, along with a **10-minute recorded video presentation** embedded on your site.

Both the website and video presentation must include:

- **Original graph or table:**
A copy of the original visualization, with proper citation of the source.
- **Analysis of the original graph or table:**
Describe the context and intent of the original visualization. Highlight its limitations and suggest areas for improvement. Address visual communication issues, missed patterns, and design flaws, and connect your suggestions to class (or other) design principles.

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- **Special efforts (if applicable):**
Note any additional work such as obtaining, cleaning, or restructuring data, or extra design/production steps taken.
- **Redesigned visualization(s):**
Present and discuss your redesigns, focusing on the patterns or insights they reveal more effectively.
- **Data storytelling component:**
Develop a clear data story based on your redesigned visualizations. Explain the narrative the data communicates, the key message for the audience, and how your design choices support that story.
- **Concluding remarks:**
Provide final reflections and possible next steps (e.g., collecting more data, fitting a different model, or creating additional plots). Producing an additional visualization will earn a bonus point.
- **Software requirement:**
All redesigns must be created using **R only**.

Minimum requirements:

For this project, you have the option to work individually or in a group of two.

- Individual:
 - Must have 1 – Original Visualization.
 - Minimum of 2 redesigns.
- Group of two:
 - Must have 1- Original Visualization with minimum of 4 redesigns.
 - **Or** 2 -Original Visualizations with minimum of 2 redesigns for each.

The video Presentation

- The video presentation should be **no longer than 10 minutes**. You will be penalized 2 points per additional minute of the video.
- The video oral presentation can differ from the written materials from the website. First, it can be less formal and more entertaining. A little tasteful humor is okay. It is okay to poke fun at poor designs (not the designers). Presentations are, hopefully, educational to classmates/viewers in terms of sharing the examples or resources and methods used for producing the redesign. (The presentation is not the forum to provide an instructional lecture).
- The presentation is to provide a positive experience for both presenters and class members. For some students this may be their first presentation. For some it may be their first presentation in English. Every student is to be treated with respect! So, please be respectful when typing your comments on the videos.
- Respectful questions and constructive comments are fine when the intent is to learn, to make the presenter look good, or to gently guide the presenter in a better direction.

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- While the video presentation does not receive its own grade, it is included in the Redesign Project rubric. If the presentation is not included, if it is in bad taste, or if it blatantly goes against class guidelines points (perhaps all) will be deducted from the final written report score.

Instructions on How to Record, Edit and Embed the Video Presentation:

Recording the video:

- You don't have to prepare a separate slide/presentation for the video. **Use your mid project Webpage as the visual.**
- Make sure to record both the webpage and you during the recording.
- All members should present part of the presentation.
- Videos should not exceed more 10 minutes.
- There are several ways to record:
 - Using OBS Studio (More Options):
 - Download OBS Studio (Free): <https://obsproject.com>
 - Tutorial 1 : <https://www.youtube.com/watch?v=bmKRSyLNEhI>
 - Tutorial 2 : <http://tinyurl.com/yzjmbvvy>
 - Using Zoom (Recommended: Easy to use):
 - Tutorial: <http://tinyurl.com/3cbs75xr>
- If you need to Edit videos before publishing, you can use the following:
 - Windows 10: Windows Video Editor (in-built)
 - Windows 11: Clipchamp (in-built)
 - Mac OS: iMovie (in-built)

Publishing the video:

- You can use **YouTube** to upload your video presentation then use the video URL to embed on your website.
 - Tutorial: <http://tinyurl.com/yup9t9kj>
 - Make sure to check these options:
 - Allow Embedding
 - Enable Comments
 - Visibility: **Unlisted** (Unlisted videos can be seen and shared by anyone with the link. Your unlisted videos won't appear in the Videos tab of your channel homepage. They won't show up in YouTube's search results.)
- Embedding videos on your Quarto Website:
 - Instructions and Codes: <https://quarto.org/docs/authoring/videos.html>

Please let you instructor know if you are having any trouble with these.

The Website

Your mid project webpage should be formatted properly and professionally. Webpage should have clear sentences and succinct paragraphs. You must include a References section at the end of the Webpage.

This is a graduate level class, so the quality of your writing should meet the requirements of academic writing. Every resource used needs to be referenced properly (preferably APA format). Failing to reference books, documents, R, R packages and codes, will result in losing points. Copy and pasted text (without quoting and fully referencing) is considered plagiarizing.

Once the webpage is written, check the spelling and grammar with available tools such as those in Word. After the initial writing, in this class it is not an honors code violation to get help in revising your paper from the university writing center. It is also okay to have a native English speaker read the materials and comment. Omit the humor and tips for classmates that may have been part of the presentation.

Asking your instructor about the suitability of a candidate bad graph or table for redesign is fine. The goal is to learn and develop professional communications skills that make you a valued resource in this culture.

Codes:

You are supposed to submit your codes as a separate Webpage on your Website. This will help your instructor to understand your approach to coding and the techniques that you used for each visualization.

Recommended: Graphic Design Guidelines

For the project, you should follow best practices in graphic design. Remember that making comparisons is at the heart of quantitative graphics. The four general guidelines are:

1. Enable accurate comparisons

- If the encoding is poor, consider a position along scale encoding.
- If position along a scale is not an option, consider upgrading a poor encoding to a good encoding.
- Adding grid lines can help.

2. Simplify appearance

- Creating small perceptual groups at one or more scales can help.
- Using easily discriminated symbols can help.
- Sorting rows or columns to put similar items close together can help.
- Proximity linking is often simpler than color linked labels.
- Using familiar colors with common names can facilitate communication.
- Explicit display of changes or differences, such as the differences between observed and fitted values, helps to show what is hard to envision.

3. Provide or increase context to support interpretation or to facilitate hypothesis generation

- Include the units of measures. Don't forget labeling and titling of the graphs!
- If the data has a spatial context adding a map can help.
- If the data is a times series adding more recent data can help.
- In an exploration in modeling context including one or more potentially related variables may be very helpful.

4. Attract and engage the reader/analyst

- Making text easier to read can help.
- Labeling a few points and/or providing annotation can draw attention.
- Colorful graphics can attract readers.
- Using interactive or dynamic graphics helps to engage the reader.
- Shiny or other R packages that use java scripts can be useful here.

There are tensions among the guidelines. Adding context can make the graphics more complicated while at the same time more meaningful. There are almost always design constraints such as the resolution of the screen or the size of the paper. Designing good graphics often involves picking the task(s) to emphasize and making compromises.

Notes:

Exact data is not required.

The original graph may not come with data in a convenient digital form. You need to track it down and use it for your analysis, but I don't require you to show the data on your slides. The uploading of the data is requested if it is in digital form.

Several suggestions were made above. For example, make sure to include the units of measure.

A graph about data is worthless if we don't know what is being represented.

In general, *layer* the information.

Make the most important information the most salient. For example, grid lines should appear in the background. Reference lines can be drawn on grid lines. Confidence interval lines can be drawn on reference lines and estimate symbols plotted on top. Methods other than overplotting can also be used to support visual layering.

Resulting designs to avoid

Don't just get a pie chart and redesign it to a bar chart! Providing multiple graphs is required to show the effort. Adding an inferential method and some analysis results to understand the data or trend is encouraged.

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Avoid using perspective 3-D bars in the improved design.

I do not particularly like 3-D perspective bar plots. Bars in front can hide bars in back. A perspective view can complicate reading bar heights against scale.

Side note: Some of the perspective weakness can be addressed. A translucent cutting plane parallel to the bar base that touches the reference scale helps to assess bar height values and to compare the bar heights. A dynamic interactive setting that supports move the reference plane can work well. I have seen tools that support large matrices of 3D bars.

Avoid use of pie charts in the improved design

I do not particularly like pie charts. They have merit because they are familiar and immediately convey that the percentages add to 100% (or fractions add to 1). Angles have good perceptual accuracy of extraction, which is not bad but not as good as position along a scale.

Mostly, I find pie charts hard to label in comparison to row-labeled dot plots that have easy-to-read labels. The labels in dot plots link by vertical position to the dots that values using position along a scale encoding. The perceptual grouping of rows can increase linking speed and accuracy.

Both designs can sort values in ascending or descending order. For pie charts, this will put the smallest value by the largest value. With pie charts including additional variables to increase context is a challenge.

Outstanding Redesigns

Some students chose to invest considerable thought, creativity, and effort in the redesign project. I give A+'s to redesigns that represent serious efforts to improve a non-trivial graph or table by adapting and applying the methods taught in class, providing multiple ways of improving graphs, and going for the best options (not the easy options!).

Grading Criteria:

Level of effort: effort is recognized in many areas.

- Selection of a non-trivial example to redesign (Converting three 4-item pie charts into 4 lines on a line plot is at best B level work.)
- Attention to appearance details
- Attention to comparability issues.
- Data gathering and preparation of data for analysis/graphics
- Development of a good, new graphics design
- Discovery of a misrepresentation
- Discovery of a mismatch between an article and the graphic
- Representing a large or complex table

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Use of concepts taught in class

- Perceptual grouping and layering to facilitate comparison
- Background grid lines, grid line labels (Avoid tick marks when there is a reasonable alternative.)
- Context for appropriate interpretation (Units of measure, legends if necessary)
- Distributional summaries
- Clean appearance:
 - No points plotted on axes
 - Easy to read labeling
 - Minimal memory burdens
 - Avoid too many font sizes. Use consistent fonts.
- Focus on the key issue (The graphics in some articles are off target)
- Quality of description
 - Indicate the nature of the dataset.
 - The bad design flaws.
 - Indicate the goals for the redesign.
 - Point out the improvements.
 - Clear labeling in the graphics (for example variable names).
 - Clarity of the text.